

Protein Discovery for Pollution Mitigation

Santiago Restrepo

WaaJacu

<https://waajacu.com/>

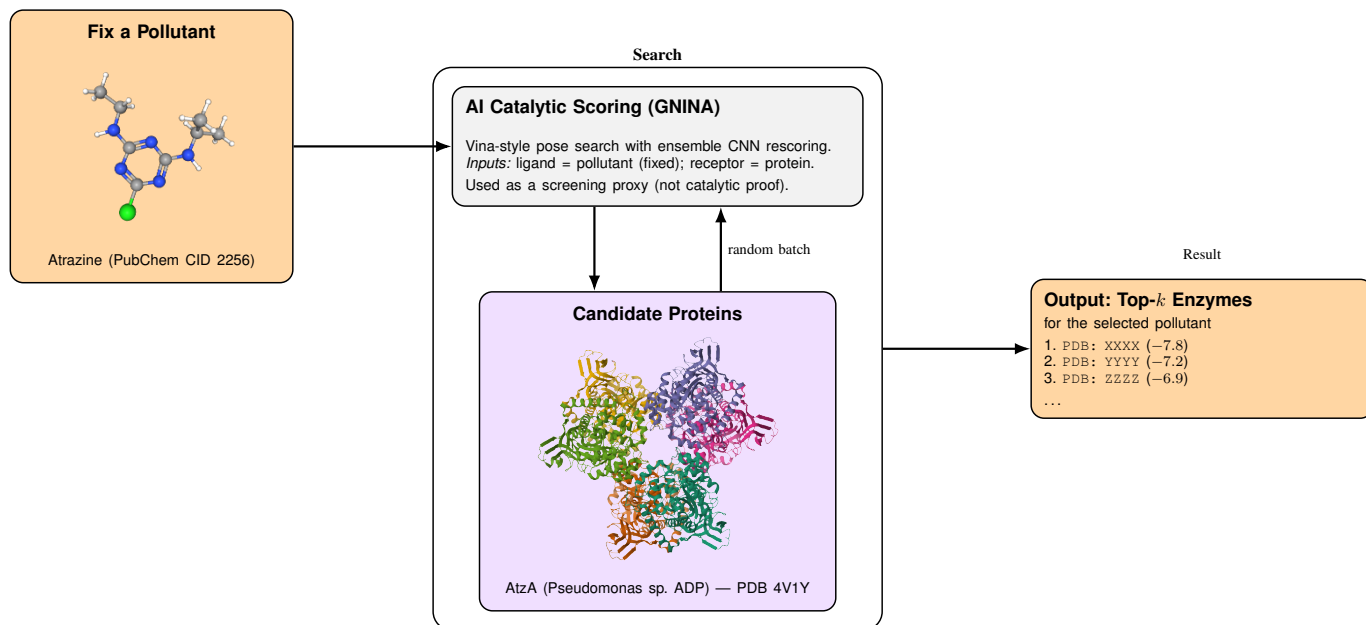


Figure 1. Pollutant-Enzyme exploration workflow diagram.

Abstract—This work presents a demonstration of how open-source tools and freely available databases can be integrated to explore possible bioremediation strategies. No new algorithms or methods are introduced. Instead, we show how existing resources can be combined into a workflow that identifies candidate protein structures with potential to interact with common pollutants. The demonstration focuses on pollutants reported in the Brahmaputra.

I. INTRODUCTION

Industrial and agricultural waste contribute significantly to water pollution in major river systems such as the Brahmaputra. While global research in protein engineering and biocatalysis continues to advance, the goal of this WaaJacu project is not to propose novel science. Rather, we illustrate how openly available computational tools and datasets can be chained together to demonstrate a possible pathway toward protein-based pollution mitigation.

II. METHODOLOGY

Inputs. *Pollutants* (SDF; <https://pubchem.ncbi.nlm.nih.gov>) and *candidate receptor proteins* (PDB; <https://www.rcsb.org>). For the demo, we *fix one pollutant* as the ligand and explore a pool of proteins via random sampling.

Search strategy. With the pollutant held fixed, we iterate over batches of proteins from PDB. Each batch is evaluated and the current top- k is updated.

Docking & AI scoring (GNINA). Protein-pollutant interaction is estimated with GNINA (<https://github.com/gnina/gnina>), using a Vina-style pose search with ensemble CNN re-scoring. The primary readout is predicted binding affinity (kcal/mol), where *more negative is better*.

Selection & output. Receptors are ranked by their best-pose affinity. The deliverable is the *ranked top- k enzymes* (PDB IDs with predicted affinities) for the fixed pollutant. Scores are used strictly as a *screening proxy* for interaction strength and do not establish catalytic viability; experimental validation is required.

III. CONCLUSION

This WaaJacu project explores protein-based pollution mitigation. It shows how open-source AI tools and public databases can be chained into a transparent, reproducible screening workflow. Framed around the Brahmaputra pollution context, it highlights how existing resources can help identify candidate enzymes for further research.

REFERENCES

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